Workbook · Hand-Calculated Examples

Fairness Metrics by Hand: A Workbook

Calculations behind the main lecture "Fairness in Financial AI"

Eight problems you can follow without a calculator — from the confusion matrix to the counterfactual verdict

The main lecture showed only the results. Here we count the same numbers from scratch.
Each problem is self-contained, in the order data → calculation → interpretation.

Counting from the confusion matrix reveals hidden gaps

FML Week 14 example — two groups of 100, each with 80 qualified and 0 unqualified approved

Cell	Group A	Group B	Meaning
TP (qualified approved)	60	40	Given a chance
FN (qualified rejected)	20	40	Denied a chance
TN (unqualified rejected)	20	20	Correctly screened out
FP (unqualified approved)	0	0	No loss

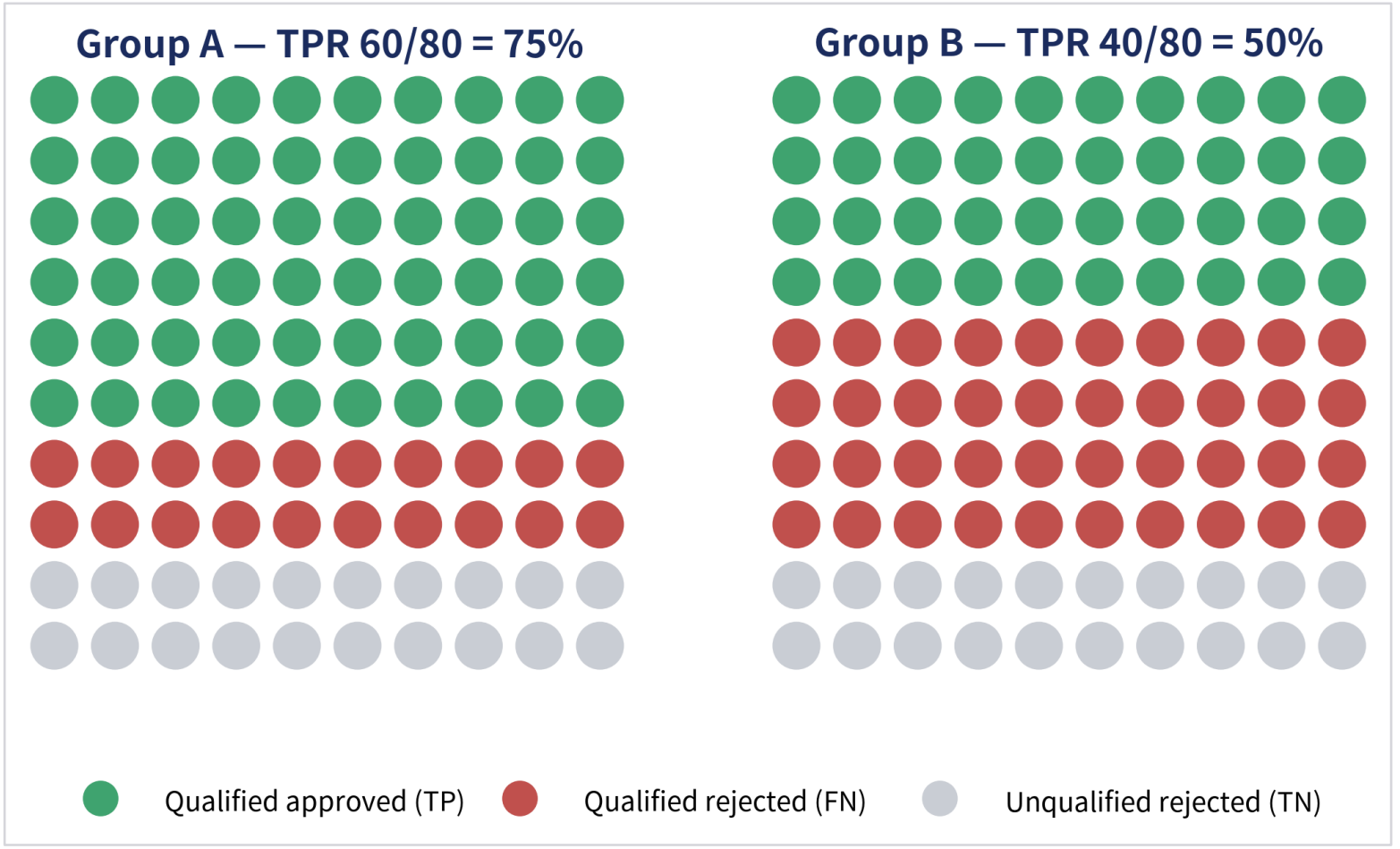
Group A: $\mathrm{TPR} = \dfrac{60}{60+20} = 75\%,$ Group B: $\mathrm{TPR} = \dfrac{40}{40+40} = 50\%$

Redraw the table as 100 dots

Green = qualified given a chance, red = qualified denied a chance, gray = unqualified correctly rejected

Count them

- A: 60 green
- A: 20 red
- B: 40 green
- B: 40 red



An average accuracy of 70% hides gaps of 20 pp and 25 pp

Compute approval rate, TPR and accuracy from the Workbook 1 table in turn

Metric	Formula	Group A	Group B	Gap
Approval rate	(TP+FP) ÷ 100	60%	40%	20 pp (demographic parity)
TPR	TP ÷ (TP+FN)	75%	50%	25 pp (equal opportunity)
Accuracy	(TP+TN) ÷ 100	80%	60%	20%p
Pooled accuracy	(80+60) ÷ 200	70%		

Interpretation

- With zero false positives it looks like a 'cautious' model, but equally qualified group B gets only half the chances.

Approving 70% of men and 50% of women fails the 4/5 rule

Source example — 700 of 1,000 men and 400 of 800 women approved

$$\mathrm{DI} = \frac{400/800}{700/1000} = \frac{0.50}{0.70} = 0.714 < 0.8$$

Step	Calculation	Result
① Approval rate	700/1,000 · 400/800	70% · 50%
② SPD	50% − 70%	−0.20
③ Both groups to 60%	600 men · 480 women	100 fewer men, 80 more women

To reach demographic parity, 100 men must be rejected and 80 more women approved, regardless of qualification.

Equal opportunity still leaves an approval-rate gap

Source example — 1,000 men and 1,000 women, 700 qualified men and 500 qualified women, TPR 90% in both groups

Item	Men	Women
Qualified approved (TPR 90%)	700 × 0.9 = 630	500 × 0.9 = 450
Overall approval rate	63%	45%
DI		45 ÷ 63 = 0.71

Interpretation

- Equal opportunity is fairness among equally qualified people. If the qualified shares differ, approval rates naturally diverge.
- Demanding both criteria at once means giving one up.

Equalized odds also matches approvals of the unqualified

Source example — men 700 repay · 300 default, women 600 repay · 400 default, TPR 90% · FPR 20%

Item	Men	Women
Repayers approved (TPR 90%)	700 × 0.9 = 630	600 × 0.9 = 540
Defaulters approved (FPR 20%)	300 × 0.2 = 60	400 × 0.2 = 80
Total approved	690 (69%)	620 (62%)
Repayers among approved (PPV)	630 ÷ 690 = 91.3%	540 ÷ 620 = 87.1%

Equal TPR and FPR still split PPV: 91.3% vs 87.1%. The next problem gives the general principle.

Equalizing PPV and FNR makes FPR differ fourfold

1,000 each, base rate A 50% · B 20%, both groups set to PPV 80% · FNR 20%

$$\mathrm{FPR}_A = \frac{0.5}{0.5} \cdot \frac{0.2}{0.8} \cdot 0.8 = 0.20, \qquad \mathrm{FPR}_B = \frac{0.2}{0.8} \cdot \frac{0.2}{0.8} \cdot 0.8 = 0.05$$

Item	Group A (500 positive)	Group B (200 positive)
TP = positives × (1−FNR)	400	160
FP = TP × (1−PPV)/PPV	100	40
FPR = FP ÷ negatives	100 ÷ 500 = 20%	40 ÷ 800 = 5%

Direct counts match the formula. With different base rates there is no way to equalize all three metrics.

Fair within departments, yet a gap appears when pooled

Berkeley 1973, applicants to six major departments — 2,691 men, 1,835 women (Bickel et al. 1975)

Department	Men applied · admit rate	Women applied · admit rate	Difference
A	825 · 62%	108 · 82%	Women +20 pp
B	560 · 63%	25 · 68%	Women +5 pp
C	325 · 37%	593 · 34%	Men +3 pp
E	191 · 28%	393 · 24%	Men +4 pp
Six departments total	~1,198 admitted · 44.5%	~557 admitted · 30.3%	Men +14 pp

Women applied mostly to low-admission departments C and E, men to high-admission A and B. Department choice creates the aggregate gap.

Reweighting makes repayment shares look equal across groups

1,000 people — A 600 (420 repay), B 400 (200 repay). Overall repayment share 62%

$$w_{A,1} = \frac{0.6 \times 0.62}{0.42} = 0.886 \qquad w_{B,1} = \frac{0.4 \times 0.62}{0.20} = 1.24$$

Cell	Original count	Weight	Weighted count
A · repay	420	0.886	372
A · default	180	1.267	228
B · repay	200	1.240	248
B · default	200	0.760	152

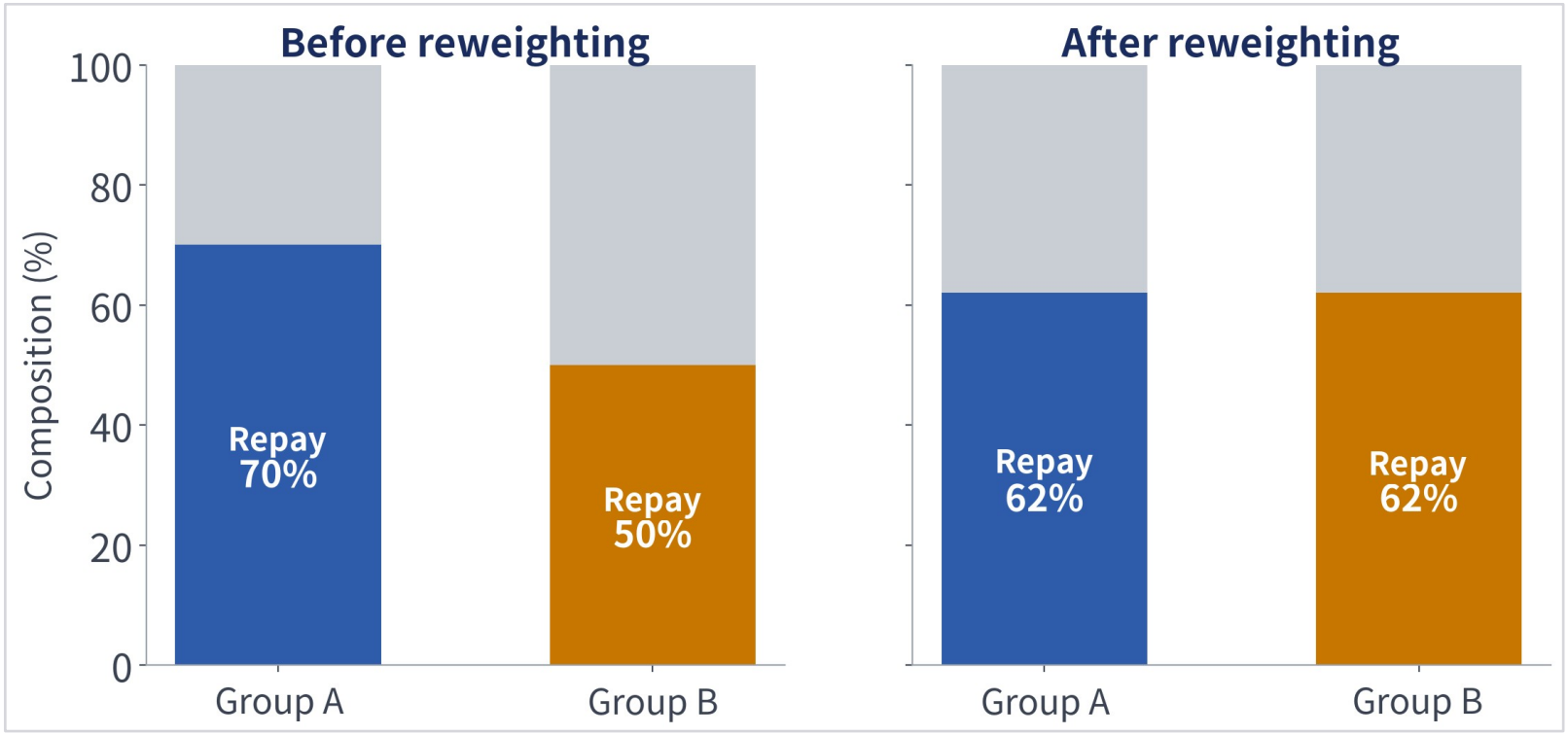
After weighting, A is 372 ÷ 600 and B is 248 ÷ 400: both groups have a 62% repayment share.

Compare repayment shares before and after weighting as bars

Head counts stay the same; only each cell's weight in training changes

Check

- A 70% → 62%
- B 50% → 62%
- Overall 62% preserved



Counterfactual calculation: same ability, attribute swapped, judged again

Toy causal model — if protected attribute A=1, income is formed 1 lower. U is unobserved ability

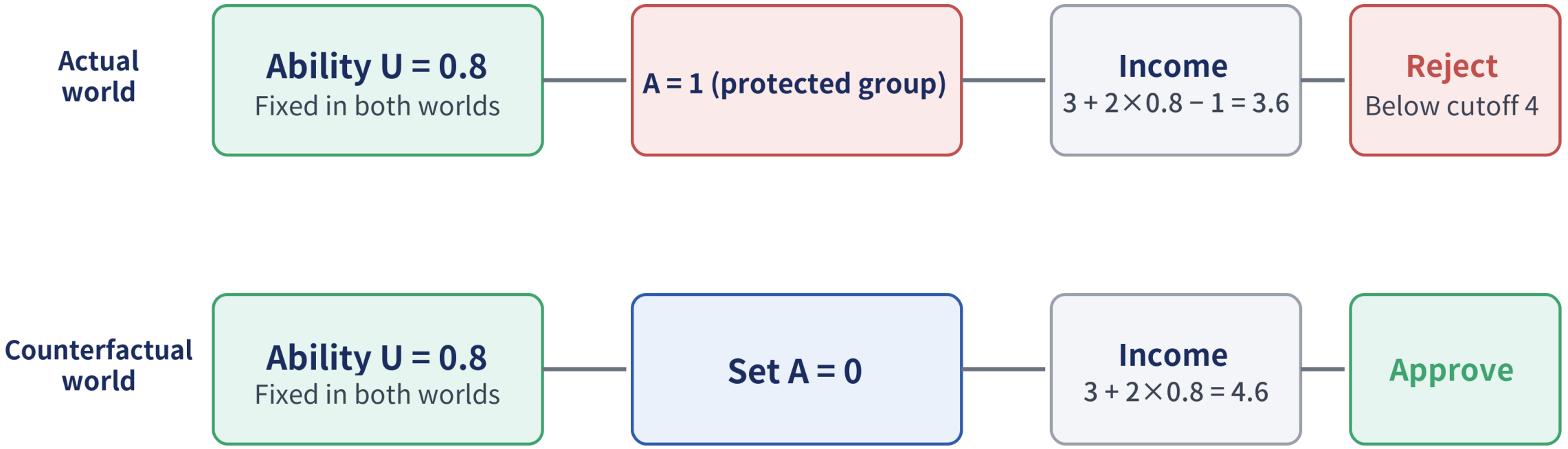
$$\text{Income} = 3 + 2U - 1 \cdot A, \qquad \text{Approve} \iff \text{Income} \geq 4$$

Step	Applicant Gap (A=1)	Applicant Eul (A=1)
Observed income	4.5 → approve	3.6 → reject
Back out ability U=(income−3+1)/2	1.25	0.80
Income had A been 0	3+2.5 = 5.5 → approve	3+1.6 = 4.6 → approve
Verdict	Counterfactually fair	Counterfactually unfair

With the same ability, Eul would have been approved in the other group. Such cases go to human review.

Place applicant Eul's two worlds side by side

Holding U at 0.8 and changing only A from 1 to 0 raises income from 3.6 to 4.6 and flips the decision



Same person (U fixed), only the attribute changed, and the decision flipped → counterfactually unfair, sent to human review

Key numbers of the eight problems

Problem	Key result	Lesson
1 Confusion matrix	Gaps 20 pp · 25 pp, average 70%	Visible only when measured by group
2 DI	0.714	Fails the 4/5 rule
3 Equal opportunity	Approval 63% vs 45%	Equal opportunity ≠ demographic parity
4 Equalized odds	PPV 91.3% vs 87.1%	Equal error rates ≠ predictive parity
5 Impossibility	FPR 20% vs 5%	Different base rates conflict
6 Simpson's paradox	44.5% vs 30.3%	Aggregation creates gaps
7 Reweighting	62% in both groups	Mitigation at the data stage
8 Counterfactual	Only Eul's verdict changes	Check at the individual level